Rodrigo Aragao

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EDUCATION

Drexel University | Philadelphia, PA | September 2021 – June 2025
Bachelor of Science in Computer Engineering, Minor in Neuroscience

EXPERIENCE

JUNIOR AI ENGINEER (REMOTE) | FACILIT TECNOLOGIA (GovTech) | NOVEMBER 2025 – PRESENT

• Designed and deployed AI workflows in n8n that ingest PDFs, Excel files, and emails into vector database, enabling an internal AI copilot to answer employee questions about contracts, dashboards, and client data in seconds instead of manual searching
• Implemented RAG pipeline end to end (chunking, embeddings, metadata logging, vector search, and answer generation), improving relevance of retrieved documents for GenAI queries and reducing hallucinations in internal tools
• Optimized document processing by batching ingestion and embeddings jobs, cutting retrieval latency by about 40% and giving sales and operations teams faster access to contract terms and KPIs for decision-making
• Engineered a high-availability AI Support Assistant using FastAPI, managing the full deployment lifecycle via Docker and Coolify
• Developing an MCP (Model Context Protocol) server in Python to enable LLM integration with the enterprise planning platform, allowing natural language queries over organizational goals, budgets, and project status for government clients

DATA ANALYTICS ENGINEER CO-OP | GS1 US | APRIL 2024 – SEPTEMBER 2024

• Designed and maintained ETL data pipelines in Python and SQL to clean, validate and transform million-row business datasets
• Developed PySpark notebooks and SQL views in Azure Databricks for barcode prefix analysis used across global trade datasets
• Increased data transformation efficiency by ~35% by optimizing PySpark workflows and SQL logic, reducing pipeline runtime and improving downstream report freshness
• Integrated and modeled cloud data sources into Power BI dashboards adopted by analytics and product teams

UNDERGRADUATE RESEARCHER, DATA SCIENCE | DREXEL UNIVERSITY, COGNEW LAB | SEPTEMBER 2023 – APRIL 2024

• Developed an automated MATLAB pipeline for multi-subject preprocessing and feature extraction across 20+ multi-channel EEG datasets, transforming raw brain data into structured metrics for downstream analysis
• Designed and tuned SVM model in Python to predict reaction times, improving prediction accuracy from 50% baseline to 63% through grid search, hyperparameter optimization, and k-fold cross validation
• Automated full ML pipeline, from signal processing to model training and evaluation, advancing research on cognitive attention

DATA SCIENCE & VISUALIZATION CO-OP | FULTON BANK, IT CHIEF DATA OFFICE | APRIL 2023 – SEPTEMBER 2023

• Created Python scripts to perform data quality profiling (completeness, uniqueness, and consistency) in Lending & Risk datasets
• Built and enhanced Power BI reports, including Delinquency and Risk Rating dashboards for key business insights
• Supported data governance by tracing lineage, creating validation checks, and enhancing reproducibility across historical datasets

PROJECTS

HBO's Succession TV Series AI Chatbot – Machine Learning | August 2024 – February 2025

• Extracted text data from scripts, isolating character's lines with custom heuristics and performing exploratory data analysis.
• Built a FAISS similarity index on language embeddings to retrieve contextually relevant quotes, and fine-tuned GPT LLM using targeted RAG to emulate each character's unique style and tone
• Launched web app with character selection and chat history, showing end-to-end AI prototyping and user-facing deployment

Litema App – Recommender System Hub | January 2026 – Present

• Developing a cross-media recommendation platform using Streamlit and Supabase, featuring a Universal Catalog shared across users and a private User-Item Bridge architecture to manage personal libraries and "Seen" lists
• Orchestrated a data pipeline using OpenAI (GPT-4o-mini) to transform raw metadata from TMDB and Hardcover APIs into high-density "Conceptual Signatures," moving beyond keyword search to capture the thematic and philosophical subtext of media
• Integrated Sentence-Transformers to generate 384-dimensional embeddings for semantic similarity, leveraging PostgreSQL/pgvector and Cosine Similarity to provide recommendations that outperform standard LLM-based linguistic guesses

SKILLS

Languages & Tools: Python, SQL, MATLAB, C, C++, PySpark, Git, Linux, FastAPI, htttpx, Excel, Fast MCP, GraphQL, REST API
Machine Learning: PyTorch, TensorFlow, HuggingFace Transformers, Scikit-learn, Sentiment Analysis
RAG: n8n, Pinecone Vector Database, Gemini File Store, Chunking, Embedding, Retrieval, Session/Context Management
Data Engineering: Azure Databricks, Azure Synapse, ETL, SQL modeling, Power BI, Databricks Notebooks, Pandas
Other: Streamlit, Tkinter, Raspberry Pi, JSON/CSV pipelines, Docker, CI/CD, Coolify, Supabase, PostgreSQL
Languages: English (Bilingual), Portuguese (Native), Spanish (Intermediate)